

```
acknowledged: true,
insertedIds: {
  '0': 1,
  '1': 2,
  '2': 3,
  '3': 4,
  '4': 5,
  '5': 6,
  '6': 7
}
}
```

```
db.products.find(
  {
    price: {
      $exists: true
    }
  },
  {
    name: 1,
    price: 1
  }
)
```

```
{ "_id" : 1, "name" : "xPhone", "price" : 799 }
{ "_id" : 2, "name" : "xTablet", "price" : 899 }
{ "_id" : 3, "name" : "SmartTablet", "price" : 899 }
{ "_id" : 4, "name" : "SmartPad", "price" : 699 }
{ "_id" : 5, "name" : "SmartPhone", "price" : 599 }
{ "_id" : 7, "name" : "xReader", "price" : null }
```

```
db.products.find({
  price: {
    $exists: true,
    $gt: 699
  }
}, {
  name: 1,
  price: 1
});
```

```
{ "_id" : 1, "name" : "xPhone", "price" : 799 }
{ "_id" : 2, "name" : "xTablet", "price" : 899 }
```

```
{ "_id" : 3, "name" : "SmartTablet", "price" : 899 }
```

```
db.products.find({  
  price: {  
    $exists: false  
  }  
}, {  
  name: 1,  
  price: 1  
});
```

```
{ "_id" : 6, "name" : "xWidget" }  
db.products.insertMany([  
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":  
ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :  
2.66 }, "color":["white","black"],"storage":[64,128,256]},  
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":  
ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :  
3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},  
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":  
ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :  
3.66 }, "color":["blue"],"storage":[16,64,128]},  
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":  
ISODate("2020-05-14"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :  
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},  
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":  
ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :  
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}  
])
```

```
{
```

```
db.products.insertMany([  
  { "_id" : 1, "name" : "xPhone", "price" : "799", "releaseDate" :  
ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :  
2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },  
  { "_id" : 2, "name" : "xTablet", "price" : NumberInt(899), "releaseDate" :  
ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :  
3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },  
  { "_id" : 3, "name" : "SmartTablet", "price" : NumberLong(899),  
"releaseDate" : ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12,  
"screen" : 9.7, "cpu" : 3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },  
  { "_id" : 4, "name" : "SmartPad", "price" : [599, 699, 799], "releaseDate" :  
ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :  
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,  
1024 ] },  
  { "_id" : 5, "name" : "SmartPhone", "price" : ["599",699], "releaseDate" :
```

```
ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
    { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
])
```

```
db.products.find({
  price: {
    $type: "string"
  }
}, {
  name: 1,
  price: 1
})
```

```
{ "_id" : 1, "name" : "xPhone", "price" : "799" }
{ "_id" : 5, "name" : "SmartPhone", "price" : [ "599", 699 ] }
```

```
db.products.find({
  price: {
    $type: 2
  }
}, {
  name: 1,
  price: 1
})
```

```
db.products.find({
  price: {
    $type: "number"
  }
}, {
  name: 1,
  price: 1
})
```

```
{ "_id" : 2, "name" : "xTablet", "price" : 899 }
{ "_id" : 3, "name" : "SmartTablet", "price" : NumberLong(899) }
{ "_id" : 4, "name" : "SmartPad", "price" : [ 599, 699, 799 ] }
{ "_id" : 5, "name" : "SmartPhone", "price" : [ "599", 699 ] }
```

```
db.products.find({
  price: {
```

```

        $type: "array"
    }
}, {
    name: 1,
    price: 1
})

```

```

{ "_id" : 4, "name" : "SmartPad", "price" : [ 599, 699, 799 ] }
{ "_id" : 5, "name" : "SmartPhone", "price" : [ "599", 699 ] }

```

```

db.products.find({
    price: {
        $type: ["number", "string"]
    }
}, {
    name: 1,
    price: 1
})

```

```

{ "_id" : 1, "name" : "xPhone", "price" : "799" }
{ "_id" : 2, "name" : "xTablet", "price" : 899 }
{ "_id" : 3, "name" : "SmartTablet", "price" : NumberLong(899) }
{ "_id" : 4, "name" : "SmartPad", "price" : [ 599, 699, 799 ] }
{ "_id" : 5, "name" : "SmartPhone", "price" : [ "599", 699 ] }

```

```

db.products.insertMany([
    { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
    ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
    2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
    { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
    ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
    3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
    { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
    ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
    3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
    { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
    ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
    1024 ] },
    { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
    ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
    { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
    3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
])

```

```

db.products.find({
  color: {
    $size: 2
  }
}, {
  name: 1,
  color: 1
})

[
  {
    _id: 1,
    name: "xPhone",
    color: [ "white", "black" ]
  }
]

```

```

db.products.find({
  $or: [{
    color: {
      $size: 1
    }
  },
  {
    color: {
      $size: 2
    }
  }
]
}, {
  name: 1,
  color: 1
})

[
  {
    _id: 1,
    name: "xPhone",
    color: [ "white", "black" ]
  },
  {
    _id: 3,
    name: "SmartTablet",
    color: [ "blue" ]
  },
  {
    _id: 6,

```

```
    name: "xWidget",
    color: [ "black" ]
  }
]
```

```
db.products.find({
  color: {
    $all: ["black", "white"]
  }
}, {
  name: 1,
  color: 1
})
```

```
[
  {
    _id: 1,
    name: "xPhone",
    color: [ "white", "black" ]
  },
  {
    _id: 2,
    name: "xTablet",
    color: [ "white", "black", "purple" ]
  }
]
```

```
db.products.find({
  $and: [
    {color: "black"},
    {color: "white"}
  ]
}, {
  name: 1,
  color: 1
})
```

```
[
  {
    _id: 1,
    name: "xPhone",
    color: [ "white", "black" ]
  },
  {
    _id: 2,
    name: "xTablet",
    color: [ "white", "black", "purple" ]
  }
]
```

```
}  
]
```

```
db.products.find(  
  storage: {  
    $elemMatch: {  
      $lt: 128  
    }  
  }  
}, {  
  name: 1,  
  storage: 1  
});
```

```
[  
  {  
    _id: 1,  
    name: "xPhone",  
    storage: [ 64, 128, 256 ]  
  },  
  {  
    _id: 3,  
    name: "SmartTablet",  
    storage: [ 16, 64, 128 ]  
  }  
]
```

```
db.products.insertMany([  
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :  
    ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :  
    2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },  
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :  
    ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :  
    3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },  
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :  
    ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :  
    3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },  
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :  
    ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :  
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,  
    1024 ] },  
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :  
    ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :  
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },  
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
```

```
3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] },
  { "_id" : 7, "name" : "xReader", "price" : null, "spec" : { "ram" : 64, "screen" :
6.7, "cpu" : 3.66 }, "color" : [ "black", "white" ], "storage" : [ 128 ] }
])
```

```
db.products.find({
  'price': {
    $exists: 1
  }
}, {
  name: 1,
  price: 1
}).sort({
  price: 1
})
```

```
[
  {
    _id: 7,
    name: "xReader",
    price: null
  },
  {
    _id: 5,
    name: "SmartPhone",
    price: 599
  },
  {
    _id: 4,
    name: "SmartPad",
    price: 699
  },
  {
    _id: 1,
    name: "xPhone",
    price: 799
  },
  {
    _id: 2,
    name: "xTablet",
    price: 899
  },
  {
    _id: 3,
    name: "SmartTablet",
    price: 899
  }
]
```


}
]